

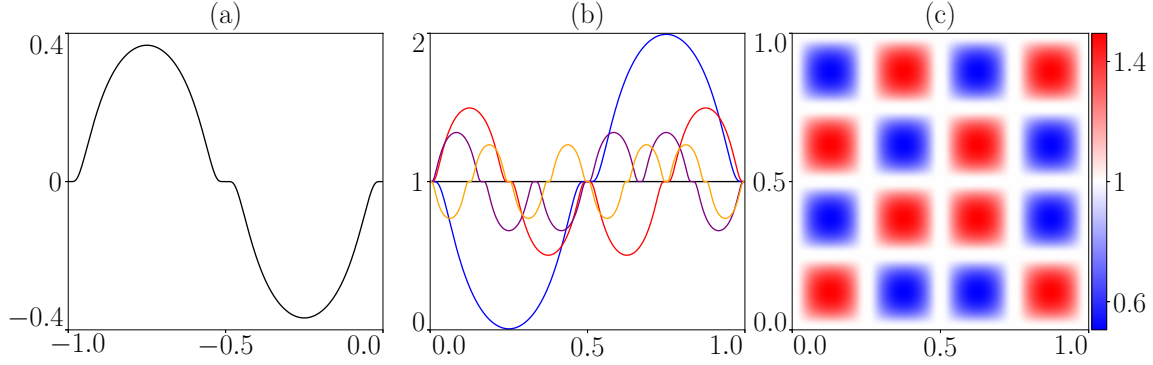


Figure 2: (a) Function G , (b) 1-dimensional uniform distribution with 0, 1, 2, 3 and 4 perturbations, (c) 2-dimensional uniform distribution with 2 perturbations.

test with different parameters, except when we compare using either a wild bootstrap or permutations, in which case we use the same samples. All our experiments are reproducible and our code is available here.

5.5 Power experiments on synthetic data

As explained in Appendix D, a lower bound on the minimax rate of testing over the Sobolev ball $\mathcal{S}_d^s(R)$ can be obtained by considering a d -dimensional uniform distribution and a perturbed version of it with $P \in \mathbb{N} \setminus \{0\}$ perturbations. As presented in Equation (19), the latter has density

$$f_\theta(u) := \mathbb{1}_{[0,1]^d}(u) + c_d P^{-s} \sum_{\nu \in \{1, \dots, P\}^d} \theta_\nu \prod_{i=1}^d G(Pu_i - \nu_i), \quad u \in \mathbb{R}^d \quad (17)$$

where $\theta = (\theta_\nu)_{\nu \in \{1, \dots, P\}^d} \in \{-1, 1\}^{P^d}$, that is, θ is a vector of length P^d with entries either -1 or 1 , and it is indexed by the P^d d -dimensional elements of $\{1, \dots, P\}^d$, and

$$G(u) := \exp\left(-\frac{1}{1 - (4u + 3)^2}\right) \mathbb{1}_{(-1, -\frac{1}{2})}(u) - \exp\left(-\frac{1}{1 - (4u + 1)^2}\right) \mathbb{1}_{(-\frac{1}{2}, 0)}(u), \quad u \in \mathbb{R}.$$

We have added a scaling factor c_d to emphasize the effect of the perturbations, in our experiments we use $c_1 = 2.7$ and $c_2 = 7.3$. Those values were chosen to ensure that the densities with one perturbation remain positive on $[0, 1]^d$. The uniform density with P perturbations for $P = 0, 1, 2, 3, 4$ when $d = 1$ and for $P = 2$ when $d = 2$, as well as the function G , are plotted in Figure 2. As shown by Li and Yuan (2019), for P large enough, the difference between the uniform density and the perturbed uniform density lies in the Sobolev ball $\mathcal{S}_d^s(R)$ for some $R > 0$. In our experiments, we choose the smoothness parameter of the perturbed uniform density defined in Equation (17) to be equal to $s = 1$. For each of the 500 repetitions used to estimate the power of a test, we sample uniformly a new value of the parameter $\theta \in \{-1, 1\}^{P^d}$ for the perturbed uniform density.

In Figure 3, we consider testing $n = 500$ samples drawn from the 1-dimensional uniform distribution against $m = 500$ samples drawn from a 1-dimensional uniform distribution

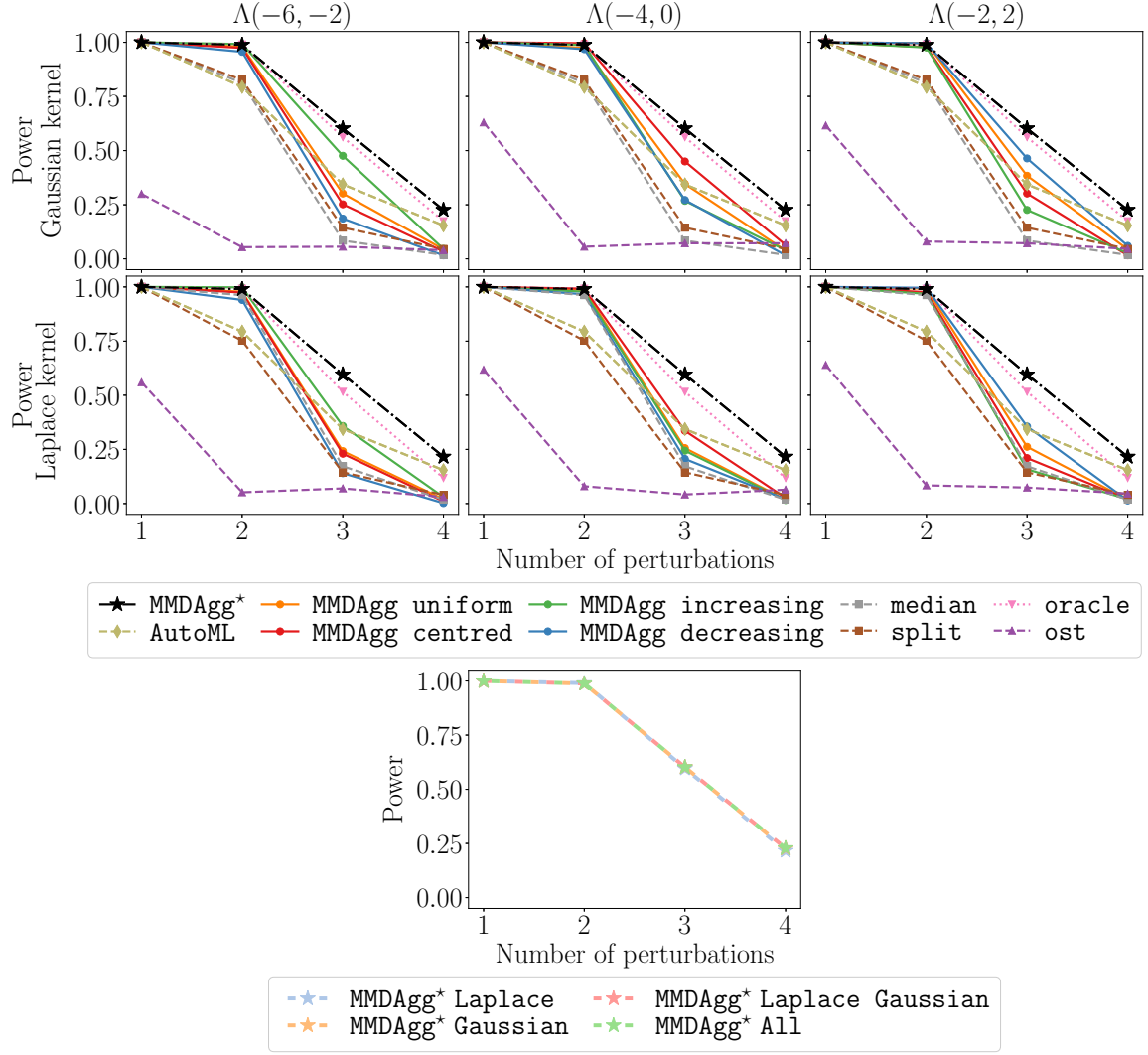


Figure 3: Power experiments with 1-dimensional perturbed uniform distributions using sample sizes $m = n = 500$ with a wild bootstrap.

with $P = 1, 2, 3, 4$ perturbations. We consider the same setting but in two dimensions with sample sizes $m = n = 2000$ with up to three perturbations in Figure 4. For the MMDAgg tests of Section 5.1 and for the `ost` test, we use the collections of bandwidths $\Lambda(-6, -2)$, $\Lambda(-4, 0)$ and $\Lambda(-2, 2)$ as defined in Equation (15). As the number of perturbations increases, it becomes harder to distinguish the two distributions, this translates into a decrease in power for all the tests. Even though we consider more samples for the 2-dimensional case, the performance of all the tests degrades significantly with dimension as detecting the perturbations becomes considerably more challenging.

For all the settings considered in Figures 3 and 4, we observe that `MMDAgg*` always performs the best, with power slightly higher than the one of `oracle` which has access to extra data to select an optimal bandwidth. All other tests achieve significantly lower test

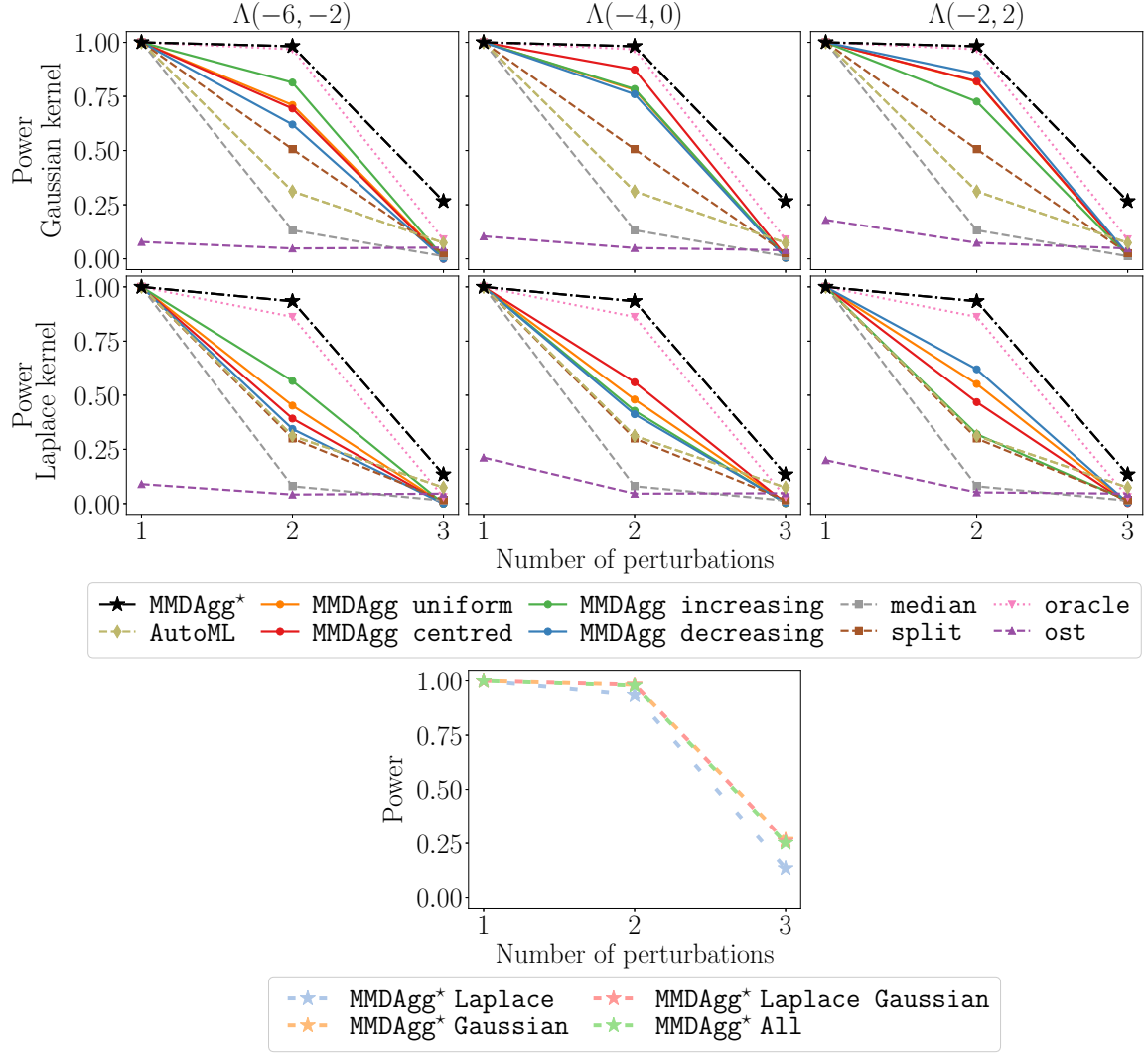


Figure 4: Power experiments with 2-dimensional perturbed uniform distributions using sample sizes $m = n = 2000$ with a wild bootstrap.

power. This validates our theoretical results that our aggregated MMDAgg test is minimax optimal in settings such as the one considered in this experiment where the difference in densities lies in a Sobolev ball of unknown smoothness. In particular, with its adaptive collection of bandwidths, MMDAgg* obtains higher power than the four other aggregated tests with specifically chosen collections. In the 1-dimensional case, AutoML performs similarly to our four aggregated tests, while in the 2-dimensional case with two perturbations it obtains much lower power. Our four tests with different weighting strategies outperform the three other tests median, split and ost in most settings, and always at least match the power of the best of those three. The ost test, which is restricted to using the linear-time MMD estimate, obtains very low power compared to all the other tests using the quadratic-time estimate.